

Assignment: Product Inventory Analysis for Apex Global Warehouse

Overview

In this assignment, you will apply your SQL and data analysis skills to a real-world dataset. You will act as a Data Analyst for a large warehouse business, using BigQuery to answer critical business questions that will drive the company's sales strategy.

Part 1: Instructions & Environment Setup

Before you begin, you must set up your BigQuery environment. This assignment is designed to be Sandbox friendly, meaning you can complete it without a credit card or incurring charges.

1. **Account Setup:** Follow the instructions in your GDA course under the section "Set up your BigQuery account for more information".
2. **Data Upload:**
 - Download the [amazon.csv](#) file.
 - Follow the steps in the GDA section "Optional: Upload the store transactions dataset to BigQuery" to upload this CSV.
3. *****Important Table Details***:**
 - Name your dataset: warehouse_analysis
 - Name your table: amazon_products
 - Ensure you check the "Auto-detect Schema" box during upload.
 - Your Schema should look like [this](#)

Part 2: The Business Story

The Scenario: You have just been hired as a Junior Data Analyst at Apex Global Warehouse, a major distribution hub for top-tier electronics and home goods. The company is preparing for its Quarterly Inventory Review, and the CEO needs to know which product lines are driving the most value and which ones have stalled.

- While the finance team has already cleaned the currency symbols for you, the data still has "traps" that could ruin your report:
- Corrupted Ratings: Some customer reviews didn't sync correctly, resulting in strange characters like | instead of a number.
- Complex Categories: Products are grouped into long strings (e.g., Electronics|HomeAudio|Accessories). You'll need to figure out how to find specific keywords within those strings.
- Massive Volume: With thousands of rows, you can't just scroll through the data, instead, you must use SQL to summarize the big picture.

Your Mission: Use BigQuery to navigate these issues and extract 10 key insights. Your final report will help the CEO decide which "Premium" items to promote and which "Budget" items to clear out of the warehouse.

Part 3: Business Questions & SQL Answers

1. Quick Stock Check

Question: Management wants a quick look at our inventory. Show the *product_name*, *category*, and *actual_price* for the first 10 products.

Tip: Think about how to "cut off" the results so you don't see the entire warehouse list at once.

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```

1 Select product_name,category,actual_price
2 From tkh-project-485015.warehouse_analysis.amazon_products
3 LIMIT 10

```

✓
Query completed

Using on-demand processing quota

Query results
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Row	product_name ▼	category ▼	actual_price ▼
1	Khaitan ORFin Fan heater for Ho...	Home&Kitchen Heating,Coolin g&AirQuality RoomHeaters Fa nHeaters	2495.0
2	Personal Size Blender, Portable Blender, Battery Powered USB Blender, with	Home&Kitchen Kitchen&Home Appliances SmallKitchenAppli ances HandBlenders	1499.0

Results per page:
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1 – 10 of 10
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2. Identifying High-Value Items

Question: Find all products where the *actual_price* is greater than 5,000. Show the name and the price.

Tip: Use a filter to only look for items that meet a specific financial threshold.

The screenshot shows the Google Cloud BigQuery console interface. The left sidebar contains navigation options: Overview, Studio, Agents, Pipelines & Integration, Data transfers, Dataform, Scheduled queries, Scheduling, Governance, Sharing (Analytics Hub), Policy tags, Metadata curation, Administration, Monitoring, Jobs explorer, Capacity management, Partner Center, Settings, and Release Notes. The main area displays a query titled "Identifying High-Value Items" with the following SQL code:

```
1 SELECT product_name, actual_price
2 FROM tkh-project-485015.warehouse_analysis.amazon_products
3 WHERE actual_price > 5000
```

Below the query editor, a status bar indicates "Query completed". The "Query results" section shows a table with two columns: "product_name" and "actual_price". The table contains two rows of data:

Row	product_name	actual_price
1	PTron Newly Launched Force X10 Bluetooth Calling Smartwatch with 1.7" Full Touch Color Display, Real Heart Rate Monitor, SpO2,	5999.0
2	PTron Newly Launched Force X10 Bluetooth Calling Smartwatch with 1.7" Full	5999.0

At the bottom right, the console indicates "Results per page: 50" and "1 - 50 of 321".

3. The Discount Impact

Question: Calculate the total amount saved (Actual Price - Discounted Price) for each product. Call this new column *total_savings*. (is this AS)

Tip: You can perform basic subtraction between columns and then organize the results to see the biggest savers first.

The screenshot shows the Google Cloud BigQuery console interface. The left sidebar contains navigation options: Overview, Studio, Agents, Pipelines & Integration, Data transfers, Dataform, Scheduled queries, Scheduling, Governance, Sharing (Analytics Hub), Policy tags, Metadata curation, Administration, Monitoring, Jobs explorer, Capacity management, Partner Center, Settings, and Release Notes. The main area displays a query titled 'The Discount Impact' with the following SQL code:

```
1 SELECT product_name, actual_price - discounted_price AS total_savings
2 FROM tkh-project-485015.warehouse_analysis.amazon_products
```

Below the query, a status message indicates: 'This query will process 209.67 KB when run. Using on-demand processing quota'. The 'Query results' section shows a table with the following data:

Row	product_name	total_savings
1	Khaitan ORFin Fan heater for H...	1196.0
2	Personal Size Blender, Portable Blender, Battery Powered USB Blender, with Four Blades, Mini Blender Travel Bottle for Juice, Shakes,	830.0
3	Green Tales Heat Seal Mini Food Sealer-Impulse Machine	139.0

At the bottom right, it shows 'Results per page: 50' and '1 - 50 of 1465'.

4. Inventory Diversity

Question: How many unique products do we have in each category? Show the category and the count.

Tip: You'll need to "bucket" your data by category and then use a function to count what is inside each bucket.

The screenshot shows the Google Cloud BigQuery console interface. The left sidebar contains navigation links for Overview, Studio, Agents, Pipelines & Integration, Governance, and Administration. The main area displays a query titled 'Inventory Diversity' with the following SQL code:

```
1 SELECT DISTINCT category, COUNT(category)
2 FROM tkh-project-485015.warehouse_analysis.amazon_products
3 GROUP BY category
```

Below the query, a message indicates: 'This query will process 107.17 KB when run. Using on-demand processing quota'. The 'Query results' section shows a table with 3 rows and 2 columns: 'category' and 'count_1'.

Row	category	count_1
1	Home&Kitchen Heating,Cooling&AirQuality RoomHeaters FanHeaters	20
2	Home&Kitchen Kitchen&HomeAppliances SmallKitchenAppliances HandBlenders	19
3	Home&Kitchen Kitchen&Home	7

At the bottom right, it says 'Results per page: 50' and '1 - 50 of 211'.

5. Customer Favorites

Question: We need a list of highly-rated products. Find products that have a *rating* higher than 4.5 and more than 10,000 *rating_count*.

Tip: Look for a way to combine two different conditions so the product must meet both requirements.

The screenshot shows the Google Cloud console interface for a BigQuery project named 'tkh-project-485015'. A query is being executed in the 'Custom...tes' tab. The query is as follows:

```
1 SELECT product_id, product_name, rating, rating_count
2 FROM tkh-project-485015.warehouse_analysis.amazon_products
3 WHERE SAFE_CAST(rating AS float64) > 4.5 AND rating_count > 10000;
```

A status message indicates: "This query will process 222.42 KB when run. Using on-demand processing quota Processing location: US".

The query results are displayed in a table with the following columns: Row, product_id, product_name, rating, and rating_count.

Row	product_id	product_name	rating	rating_count
1	B00NFD0ETQ	Logitech G402 Hyperion Fury USB Wired Gaming Mouse, 4,000 DPI, Lightweight, 8 Programmable Buttons, Compatible for PC/Mac -	4.6	10760
2	B01MQ2A86A	Logitech M331 Silent Plus Wireless Mouse, 2.4GHz with	4.6	12375

At the bottom, it shows 'Results per page: 50' and '1 - 6 of 6'.

6. Discount Strategy Analysis

Question: What is the average *discount_percentage* for products in the 'Electronics' category?

Tip: Use a math function to find the middle value of the discounts, and use a wildcard to find any category that contains the word "Electronics."

The screenshot shows the Google Cloud BigQuery console interface. On the left is a navigation menu with options like Overview, Studio, Agents, Pipelines & Integration, Data transfers, Dataform, Scheduled queries, Scheduling, Governance, Sharing (Analytics Hub), Policy tags, Metadata curation, Administration, Monitoring, Jobs explorer, Capacity management, Partner Center, Settings, and Release Notes. The main area displays a query titled "Discount Strategy Analysis" with the following SQL code:

```
1 SELECT AVG(discount_percentage), Category
2 FROM tkh-project-485015.warehouse_analysis.amazon_products
3 GROUP BY Category
4 HAVING Category LIKE '%Electronics%'
```

Below the query, a message states: "This query will process 118.61 KB when run. Using on-demand processing quota". The "Query results" section shows a table with 6 rows and 3 columns: Row, f0_ (average discount), and Category.

Row	f0_	Category
1	0.698157894736...	Electronics Wearable Technolog...
2	0.595102040816...	Electronics HomeTheater,TV&VI...
3	0.382857142857...	Electronics HomeTheater,TV&Video Televisions SmartTelevi...
4	0.582692307692...	Electronics Headphones,Earbud...
5	0.23	Electronics HomeTheater,TV&VI...
6	0.185555555555...	Electronics Mobles&Accessori...

At the bottom right, it indicates "Results per page: 50" and "1 - 50 of 65".

7. Cleaning Up the Ratings

Question: Some ratings are missing or corrupted (like the '|' character). Use a CASE statement to show the *product_name* and a new column that turns any rating of '|' into 0.0, otherwise keep the original rating.

Tip: Create a logic "switch" that checks the value of the rating and changes it only if it matches a specific "bad" character.

amazon...cts x *Untitled...ery x

Untitled query Run Save Download Share Schedule Open in More

```

1 SELECT
2   product_name,
3   CASE
4     WHEN rating = '' THEN 0.0
5     ELSE SAFE_CAST (rating AS FLOAT64)
6   END AS cleaned_rating
7 FROM tkh-project-485015.warehouse_analysis.amazon_products
8
9

```

Query completed

Using on-demand processing quota

Query results + Create conversation Save results Open in

Job information Results Visualization JSON Execution details Execution graph

Row	product_name	cleaned_rating
1	Khaitan ORFin Fan heater for H...	2.0
2	Personal Size Blender, Portable Blender, Battery Powered USB Blender, with Four Blades, Mini Blender Travel Bottle for Juice, Shakes,	2.3
3	Green Tales Heat Seal Mini	2.6

Results per page: 50 1 - 50 of 1465

8. Pricing Brackets

Question: Create a column called *price_bracket* where:

- actual_price < 500 is 'Budget'
- actual_price between 500 and 2000 is 'Mid-Range'
- actual_price > 2000 is 'Premium'

Tip: Use multiple logic branches to categorize your products based on their cost ranges.
Once do the work AS

Google Cloud TKH Project Search (/) for resources, docs, products, and more

BigQuery Set up billing to upgrade to the full BigQuery experience. [Learn more](#) Dismiss Upgrade

Overview **Studio** Agents **Preview**

Pipelines & Integration

- Data transfers
- Dataform
- Scheduled queries
- Scheduling

Governance

- Sharing (Analytics Hub)
- Policy tags
- Metadata curation

Administration

- Monitoring
- Jobs explorer
- Capacity management
- Partner Center
- Settings **Preview**
- Release Notes

Pricing Brackets Run Share Schedule Open in More Save query Download

```

1 SELECT product_name, actual_price,
2 CASE
3   WHEN actual_price < 500 THEN 'Budget'
4   WHEN actual_price BETWEEN 500 AND 2000 THEN 'Mid-Range'
5   WHEN actual_price > 2000 THEN 'Premium'
6 END AS price_bracket
7 FROM tkh-project-485015.warehouse_analysis.amazon_products

```

✓ This query will process 198.22 KB when run.
Using on-demand processing quota

Query results + Create conversation Save results Open in

Job information	Results	Visualization	JSON	Execution details	Execution graph
Row	product_name	actual_price	price_bracket		
1	Khaitan ORFin Fan heater for H...	2495.0	Premium		
2	Personal Size Blender, Portable Blender, Battery Powered USB Blender, with Four Blades, Mini Blender Travel Bottle for Juice, Shakes,	1499.0	Mid-Range		
3	Green Tales Heat Seal Mini Food Sealer-impulse Machine	300.0	Budget		

Results per page: 50 1 - 50 of 1465

9. Missing Data Audit

Question: Identify any products where the *rating_count* is missing. This is crucial for our data integrity report.

Tip: Search for a specific state of data where a cell is completely empty. NULL

The screenshot displays the Google Cloud BigQuery interface. On the left is a navigation sidebar with sections like Overview, Studio, Agents, Pipelines & Integration, Governance, and Administration. The main area shows a query titled 'Missing Data Audit' with the following SQL:

```
1 SELECT product_id, product_name, rating_count
2 FROM tkh-project-485015.warehouse_analysis.amazon_products
3 WHERE rating_count IS NULL
```

Below the query, a status message indicates: 'This query will process 215.37 KB when run. Using on-demand processing quota'. The 'Query results' section shows a table with 2 rows and 4 columns: product_id, product_name, rating_count, and an empty column. The data is as follows:

Row	product_id	product_name	rating_count
1	BOB94JPY2N	Amazon Brand - Solimo 65W Fast Charging Braided Type C to C Data Cable Suitable For All Supported Mobile Phones (1 Meter, Black)	null
2	BOBQRJ3C47	REDTECH USB-C to Lightning Cable 3.3FT, [Apple MFI Certified] Lightning to Type C	null

At the bottom right, it says 'Results per page: 50' and '1 - 2 of 2'.

10. Competitive Categories

Question: Which categories have an average rating above 4.0? Only show categories that have more than 5 products.

Tip: After you group your data, you'll need to filter the results of your math rather than the raw rows.

The screenshot shows the Google Cloud BigQuery console interface. The top navigation bar includes the Google Cloud logo, the project name 'TKH Project', and a search bar. The left sidebar contains a navigation menu with options like Overview, Studio, Agents, Pipelines & Integration, Data transfers, Dataform, Scheduled queries, Scheduling, Governance, Sharing (Analytics Hub), Policy tags, Metadata curation, Administration, Monitoring, Jobs explorer, Capacity management, Partner Center, Settings, and Release Notes. The main area displays a SQL query titled 'Competitive Categori...' with the following code:

```
1 SELECT category, AVG(SAFE_CAST(rating AS float64)), COUNT(product_id)
2 FROM tkh-project-485015.warehouse_analysis.amazon_products
3 GROUP BY category
4 HAVING AVG(SAFE_CAST(rating AS float64)) > 4.0 AND Count(product_id) > 5
```

Below the query, a status message indicates: 'This query will process 131.38 KB when run. Using on-demand processing quota'. The 'Query results' section shows a table with 3 rows and 5 columns. The columns are labeled 'Row', 'category', 'f0_', 'f1_', and an unlabeled column. The data is as follows:

Row	category	f0_	f1_	
1	Home&Kitchen(Kitchen&Home Appliances)SmallKitchenAppliances)HandBlenders	4.057894736842...		19
2	Electronics(WearableTechnolog...	4.0249999999999...		76
3	Computers&Accessories(Acce ssories&Peripherals)Cables&Accessories)Cables)USBCables	4.151931330472...		233

At the bottom right, it says 'Results per page: 50' and '1 - 44 of 44'.

List of Necessary Queries to know:

- **SELECT ... FROM:** The foundation of every query.
- **AS (Aliasing):** Giving your calculated columns a professional name.
- **WHERE vs HAVING:** Understanding when to filter raw data and when to filter grouped data.
- **GROUP BY:** The key to summarizing data by category.
- **ORDER BY DESC:** How to sort your best-selling or most-expensive items to the top.
- **CASE WHEN:** How to create "if-then" logic inside your SQL tables.
- **LIKE '%Text%':** Using wildcards to find keywords within long category strings.
- **IS NULL:** The standard way to find missing information.